1. m_0 ν

v

$$\frac{h\nu}{c} = \frac{m_0 v}{\sqrt{1-(v^2/c^2)}} \quad (1)$$

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} \quad (2)$$

(1)、(2) v

$$E = \sqrt{m_0^2 c^4 + h^2 \nu^2} \quad (3)$$

$$E_0 = h\nu + m_0 c^2 \quad (4)$$

(3)、(4)

$$E^2 - E_0^2 = (m_0^2 c^4 + h^2\nu^2) - (h\nu + m_0 c^2)^2 = -2h\nu m_0 c^2 \neq 0 \quad (5)$$

2.

W

v $h\nu$

m_0

$$h\nu \geqslant \left[\frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} - m_0 c^2\right] + W \quad (6)$$

v

$$\frac{1}{\sqrt{1-(v^2/c^2)}} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + \cdots \quad (7)$$

(7) (6)

$$h\nu \geqslant \frac{1}{2} m_0 v^2 + W \quad (8)$$

$h\nu \geqslant W$ (8)

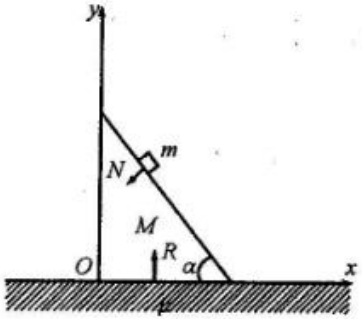
BC

Oxy a_x、a_y

x y

N R

a_M



$$ma_x = N\sin\alpha \quad (1)$$

$$ma_y = mg - N\cos\alpha \quad (2)$$

$$Ma_M = N\sin\alpha - \mu R \quad (3)$$

$$R = N\cos\alpha + Mg \quad (4)$$